Ouranos Reconstruction – RDPS-CaPA (ORRC) v1.0 / Reconstruction Ouranos – RDPS-CAPA (RORC) v1.0

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1 Background and Summary

The Canadian Surface Reanalysis (CaSR) (link) dataset is a high-resolution regional surface reanalysis developed by Environment and Climate Change Canada (ECCC). It provides an observation-based representation of surface and near-surface atmospheric conditions over North America, Central America, and nearly all of Greenland from 1968 through 2024 (Gasset et al., 2021; Khedhaouiria et al., 2026). Continuous climate monitoring and operational climate services, however, require high-resolution reanalysis data through the present day. To ensure continuity beyond the CaSR period, we produced the Ouranos Reconstruction – RDPS-CaPA (ORRC), version 1.0. The methodological framework was designed by ECCC to generate a CaSR-like dataset from the Regional Deterministic Prediction System (RDPS; link) and the Regional Deterministic Precipitation Analysis (RDPA; link), which is based on the Canadian Precipitation Analysis (CaPA) system. Ouranos implemented the end-to-end data production. ORRC provides hourly surface and near-surface meteorological fields over the same domain as CaSR v3.2 at a resolution of 0.09° (~10 km) on a rotated uniform latitude-longitude grid, over land and ocean, from 2015 to present.

ORRC's primary contribution is temporal continuity at high spatial and temporal resolution, enabling near-real-time climate monitoring and applications that compare the most recent conditions against historical baselines. This is particularly valuable for operational climate services, event monitoring, contextual analysis, and evidence-informed planning, allowing practitioners to assess recent events – such as heatwaves, heavy rainfall, or droughts – in a long-term context without waiting for official reanalysis updates.

2 Methodology

2.1 Input data

2.1.1 Regional Deterministic Prediction System (RDPS)

RDPS is ECCC's operational short-range deterministic numerical weather prediction (NWP) system for regional short-range forecasting (ECCC, 2024b). It is based on the Global Environmental Multiscale (GEM) atmospheric model and provides forecasts of atmospheric and near-surface variables at approximately 10 km horizontal resolution. Forecasts are initialized four times daily at 00, 06, 12, and 18 UTC and extend to 84 hours. The distributed regional products cover North America and adjacent oceans on a rotated latitude-longitude grid. To construct ORRC, we retain only the 6- to 12-hour forecast lead times from each RDPS cycle to minimize spin-up errors and model drift.

2.1.2 Canadian Precipitation Analysis (CaPA-RDPA)

The Canadian Precipitation Analysis (CaPA) is ECCC's system for producing gridded precipitation analyses (Fortin et al., 2028). In its operational deterministic configuration, CaPA produces the Regional Deterministic Precipitation Analysis (RDPA), hereafter referred to as CaPA-RDPA. CaPA-RDPA combines a precipitation background field from short-range RDPS forecasts with surface precipitation observations and, during liquid-precipitation conditions, radar and IMERG satellite precipitation estimates (ECCC, 2024a). The surface precipitation observations are subject to quality control procedures within the CaPA system, including temporal and spatial checks. The RDPS background and assimilated observations are combined using an optimal interpolation framework to produce gridded precipitation analyses.

Table 1 places CaPA-RDPA within the broader set of CaPA configurations. CaPA is used in both operational near-real-time products and reanalysis-based products, with differences in background field, spatial resolution, accumulation period, domain, temporal availability, and assimilated data sources. In ORRC production, the operational CaPA-RDPA 6- and 24-hour precipitation analyses are used as reference accumulations to adjust RDPS precipitation and produce a CaSR-like precipitation field. For simplicity, the RDPA 6- and 24-hour analyses used in ORRC production are referred to as CaPA-6h and CaPA-24h, respectively, throughout this document.

Table 1. Comparison of the operational CaPA-RDPA product and the online and offline CaPA components used in the CaSR reanalysis.

	Operational CaPA	CaPA in CaSR reanalysis	
	RDPA	**Online component (surface coupled)**	**Offline component (a posteriori)**
Configuration	Deterministic	Deterministic	Deterministic
Spatial resolution	10 km	10 km	10 km
Accumulation period	6 h / 24 h	6 h	24 h; hourly disaggregated
Spatial coverage	North America	North America	North America
Temporal coverage	2011–present	1968–2024	1968–2024
Surface station data	Yes	Yes	Yes + additional climate networks
Radar data	Yes (during liquid precipitation conditions)	No	No
Satellite data	Yes (during liquid precipitation conditions)	No	No

2.1.3 Data retrieval

CaPA-24h data for the full period, along with CaSR v3.2 data for a single date, were downloaded from ECCC's high-performance computer GPSC-C and converted to NetCDF format[1]. RDPS and CaPA-6h data up to 31 March 2025 were downloaded in NetCDF format from the Canadian Surface Prediction Archive (CaSPAr) at https://caspar-data.ca (Mai et al., 2020) under the ECCC Data Servers End-use Licence (link). Because of CaSPAr service unavailability, RDPS and CaPA-6h data starting from 1 April 2025 onward were downloaded from GPSC-C and converted to NetCDF.

2.2 Preprocessing

Prior to ORRC production, we perform a completeness and consistency check of the raw RDPS and CaPA data. For RDPS, we verify that each forecast cycle includes the seven hourly lead times (+6 to +12 h). For CaPA, we verify that 6- and 24-hour accumulation periods are represented by a file containing a single analysis time step. Missing files are generated and

[1] Note that the Ouranos ORRC v1.0 data treatment workflow retrieves RDPS and CaPA data in ECCC's FST file format via a GPSC-C collaborator account. Conversion to NetCDF is achieved using the fst2nc Python utilities: https://github.com/neishm/fstd2nc

populated with NaN values, and any missing variables in existing files are similarly filled with NaNs.

We regrid the RDPS and CaPA data to the CaSR v3.2 target grid (0.09° resolution) using nearest-neighbour interpolation. Although the input and target grids both have the ~10 km horizontal resolution, variations in grid definitions, including changes to the RDPS operational grid over time, necessitate this regridding step. Nearest-neighbour interpolation is used because it preserves original values and can accommodate varying grid structures.

2.3 ORRC production

The general methodology for ORRC dataset construction follows that described by ECCC researchers for the construction of CaSR-like files based on real-time operational products (Dimetrijevic, 2023). The details for the individual steps are described in the following sub-sections.

2.3.1 Temporal assembly and variables

Each RDPS forecast cycle provides hourly fields for lead times +6 to +12. ORRC is constructed by concatenating these selected segments from consecutive forecast cycles to form a continuous hourly time series. The resulting 24-hour reporting window spans from 13 UTC to 12 UTC of the following calendar day.

ORRC provides an hourly record of near-surface meteorological fields, including near-surface air temperature, dew point temperature, relative humidity, specific humidity, surface air pressure, sea level pressure, wind speed and direction, downwelling shortwave and longwave radiation at the surface, and precipitation (model precipitation and precipitation adjusted with CaPA-6h and CaPA-24h totals).

The temporal coverage of ORRC (2015–present) is determined by the availability of the underlying RDPS and CaPA inputs in the CaSPAr archive. The eastward and northward near-surface wind components contain a data gap from 11 June 2024 to 31 March 2025 because values in the CaSPAr archive during this period were erroneous. These values are therefore set to NaN, after which data retrieved directly from GPSC-C are used. The hourly near-surface wind direction retrieved from CaSPAr does not contain erroneous values during this period and therefore has no corresponding gap. However, daily mean wind direction is derived from the daily mean eastward and northward wind components using the *xclim* Python library (Bourgault et al., 2023), and consequently inherits the same gap. Users requiring these wind variables during the affected period should use CaSR v3.2 or alternative wind products.

We additionally provide daily aggregates, including daily means and daily minima/maxima for near-surface air temperature, dew point temperature, and relative humidity. Daily values are computed from the hourly record by resampling to a daily time step using standard calendar-day bins defined by the timestamps (midnight-to-midnight).

Table 2 shows the list of variables provided in ORRC, their corresponding ECCC identifiers and output frequencies.

Table 2. Variables provided in ORRC. The table lists the long name, variable name, corresponding ECCC variable identifier, and output frequency. Variables marked with an asterisk are also available in the bias-adjusted ORRC-a product (daily only).

Long name	ORRC variable name	ECCC variable identifier	Frequency
Precipitation*	pr	A_PR_SFC	Hourly + daily mean
Model precipitation	prmod	P_PR_SFC	Hourly + daily mean
1.5 metre temperature*	tas	P_TT_10000 (P_TT_1.5m in CaSR)	Hourly + daily mean/min/max
1.5 metre dewpoint temperature	tdps	P_TD_10000 (P_TD_1.5m in CaSR)	Hourly + daily mean/min/max
Near-Surface Relative Humidity (1.5m)	hurs	P_HR_10000 (P_HR_1.5m in CaSR)	Hourly + daily mean/min/max
Near-Surface Specific Humidity (1.5m)	huss	P_HU_10000 (P_HU_1.5m in CaSR)	Hourly + daily mean
Surface Downwelling Shortwave Radiation	rsds	P_FB_SFC	Hourly + daily mean
Surface Downwelling Longwave Radiation	rlds	P_FI_SFC	Hourly + daily mean
Surface Air Pressure	ps	P_P0_SFC	Hourly + daily mean
Sea Level Pressure	psl	P_PN_SFC	Hourly + daily mean
Eastward Near-Surface Wind (10m)	uas	P_UUC_10000 (P_UUC_10m in CaSR)	Hourly + daily mean
Northward Near-Surface Wind (10m)	vas	P_VVC_10000 (P_VVC_10m in CaSR)	Hourly + daily mean
Near-Surface Wind Speed (10m)	sfcWind	P_UVC_10000 (P_UVC_10m in CaSR)	Hourly + daily mean
Near-Surface Meteorological Wind Direction (10m)	winddir	P_WDC_10000 (P_WDC_10m in CaSR)	Hourly + daily mean

2.3.2 Precipitation reconstruction

To produce a CaSR-like precipitation field, RDPS hourly precipitation is adjusted using CaPA accumulations in two steps. First, cumulative RDPS precipitation is converted to hourly increments by differencing consecutive forecast times. Within each 6-hour window, the RDPS hourly increments are scaled so that their sum matches the corresponding CaPA-6h accumulation. If the RDPS 6-hour total is missing or near zero (less than 1×10^{-6} m), the CaPA-6h total is instead distributed uniformly across the six hours.

The resulting hourly fields are then concatenated into a 24-hour sequence and adjusted a second time using the CaPA-24h accumulation. When the 24-hour sum is valid, all hourly values are scaled so that the final daily total matches CaPA-24h. If the 24-hour sum is missing or near zero, the CaPA-24h total is distributed uniformly across the 24 hours. This two-step procedure ensures consistency with both the 6-hour and 24-hour CaPA accumulations while preserving the RDPS hourly precipitation distribution.

Figure 1 shows an example daily precipitation field from ORRC over the full domain, illustrating the spatial structure resolved at 0.09° resolution.

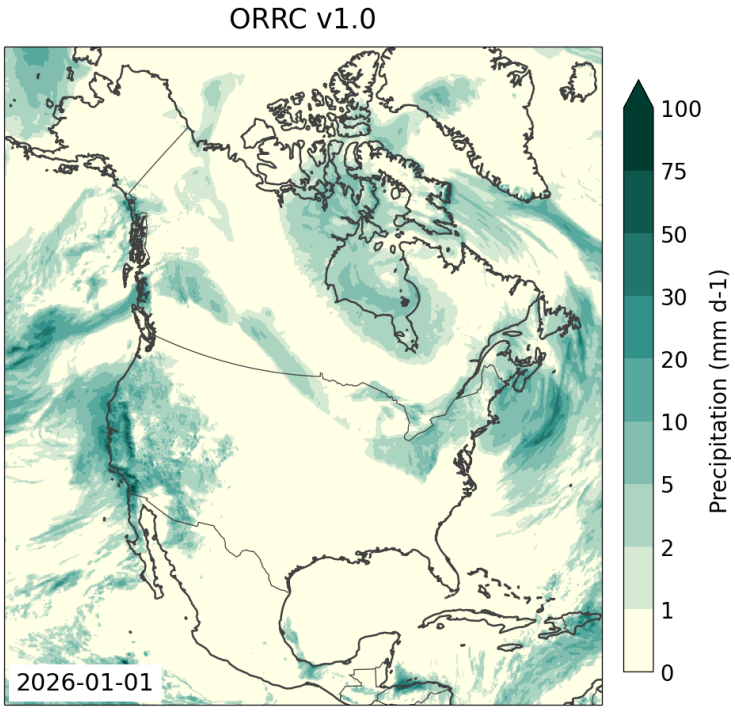


Figure 1. Daily precipitation (mm day $^{-1}$) from ORRC v1.0 over the dataset domain on 1 January 2026. The map illustrates the spatial structure at 0.09° (~10 km) resolution on the rotated latitude–longitude grid.

2.4 ORRC-a bias adjustment

To improve the continuity between ORRC and CaSR when they are used together, we produce a bias-adjusted variant of ORRC, Ouranos Reconstruction – RDPS-CaPA adjusted (ORRC-a) v1.0 for a subset of daily variables: daily mean precipitation (pr) and daily mean, minimum, and maximum 1.5 metre temperature (tas, $tasmin$, and $tasmax$, respectively). The objective of the adjustment is to reduce mean bias relative to CaSR v3.2 and allow recent conditions to be interpreted on a basis that is more consistent with the historical record. We therefore recommend ORRC-a when its daily resolution and variable coverage meet application needs. The same adjustment approach can be applied to other variables using the approach documented in the ORRC GitHub repository at https://github.com/Ouranosinc/ORRC. For applications requiring sub-daily resolution or variables for which the bias adjustment approach is not appropriate, ORRC remains a viable option, with the caveat that continuity with CaSR may vary by variable, as evaluated in the following section.

The bias adjustment is implemented using the *xsdba* Python library (Bourgault et al., 2026). We apply a day-of-year mean bias correction computed using a 31-day moving window (±15 days). For temperature variables, we apply an additive correction, and for precipitation we apply a multiplicative correction to maintain non-negative precipitation values. Because bias adjustment is applied independently to each variable, it does not preserve the physical relationships among them, which can cause the adjusted temperatures to violate the constraint $tasmin \ll tas \ll tasmax$. We therefore enforce this constraint after adjustment by swapping $tasmax$ and $tasmin$ where they are inverted, and replacing tas with the mean of $tasmin$ and $tasmax$ where it falls outside their range, computed with the *xclim* Python library (Bourgault et al., 2023), leaving all other values unchanged.

The adjustment period is 2020–2024. The start year was selected to follow the major upgrade of RDPS to version 7.0.0 in July 2019, which included a change from version 4.8 to version 5.0 of the GEM atmospheric model (ECCC, 2019), bringing the operational model configuration closer to the GEM version 5.2.1 used in CaSR v3.2 (Khedhaouiria et al., 2026). The year 2020 is the first complete calendar year following this upgrade and therefore provides the longest available period through 2024 for computing the day-of-year adjustment factors.

Figure 2 illustrates the impact of the bias adjustment on monthly mean precipitation in a representative region.

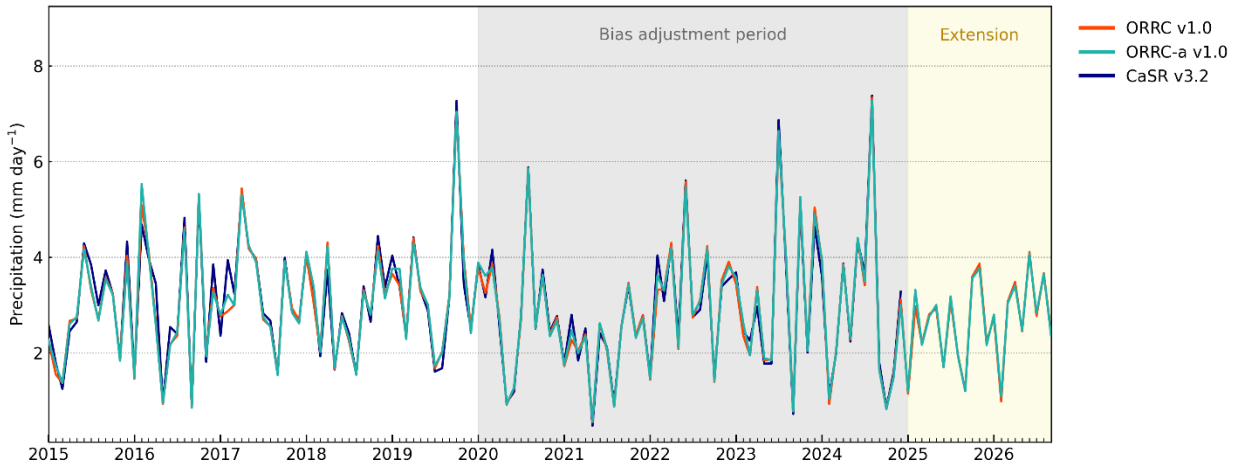


Figure 2. Monthly mean precipitation (mm day $^{-1}$) in the Montréal regional county municipality from January 2015 to August 2026, derived from ORRC v1.0, ORRC-a v1.0, and CaSR v3.2. The grey shaded area indicates the 2020-2024 period used to bias adjust ORRC and produce ORRC-a. The yellow shading indicates the period beyond the end of the available CaSR v3.2 record, covered by ORRC and ORRC-a.

2.5 Evaluation against CaSR v3.2

The evaluation of ORRC and ORRC-a relative to CaSR is conducted over 2020-2024 to coincide with the period used for the ORRC-a bias adjustment and to focus the comparison on the period following the major RDPS upgrade in 2019. The objective of this comparison is to assess the degree of similarity between the products, rather than to identify which product performs better.

The evaluation considers both temporal and spatial agreement. Regional temporal variability is assessed using Taylor diagrams computed from monthly time series spatially averaged over the selected Bukovsky regions (Bukovsky, 2011) shown in Figure 3. The Taylor diagrams (Taylor, 2001) summarize the agreement between ORRC and CaSR in terms of Pearson correlation, normalized standard deviation, and centred normalized root-mean-square error (centred nRMSE). Figure 4 illustrates this evaluation for monthly mean precipitation. All regions cluster closely around the CaSR reference point, with Pearson correlations above 0.98 and normalized standard deviations between approximately 0.90 and 1.03, indicating that ORRC closely reproduces the monthly variability of CaSR, with a modest tendency toward reduced variability. Taylor diagrams for the other evaluated variables, as well as for ORRC-a, are available in the ORRC GitHub repository.

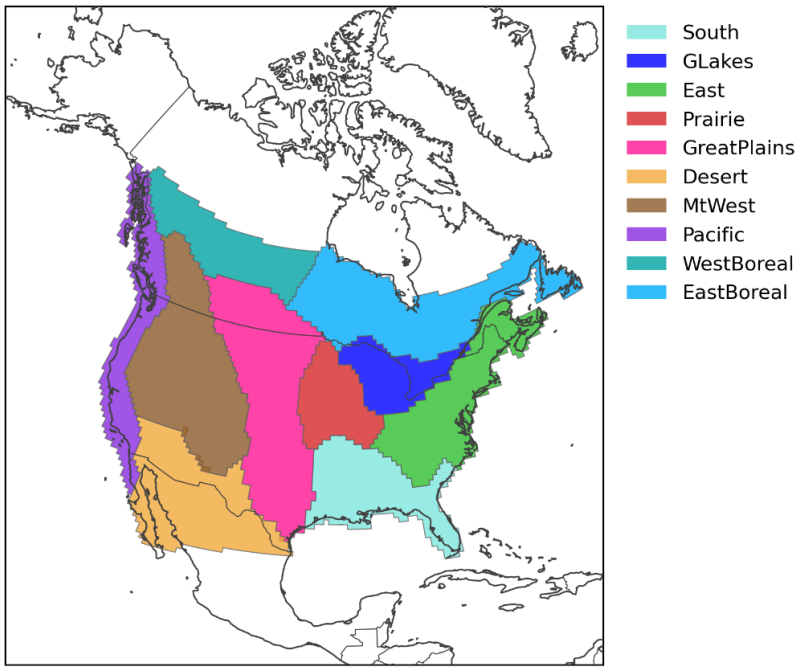


Figure 3. Bukovsky regions used for the regional evaluation of ORRC v1.0 and ORRC-a v1.0 against CaSR v3.2.

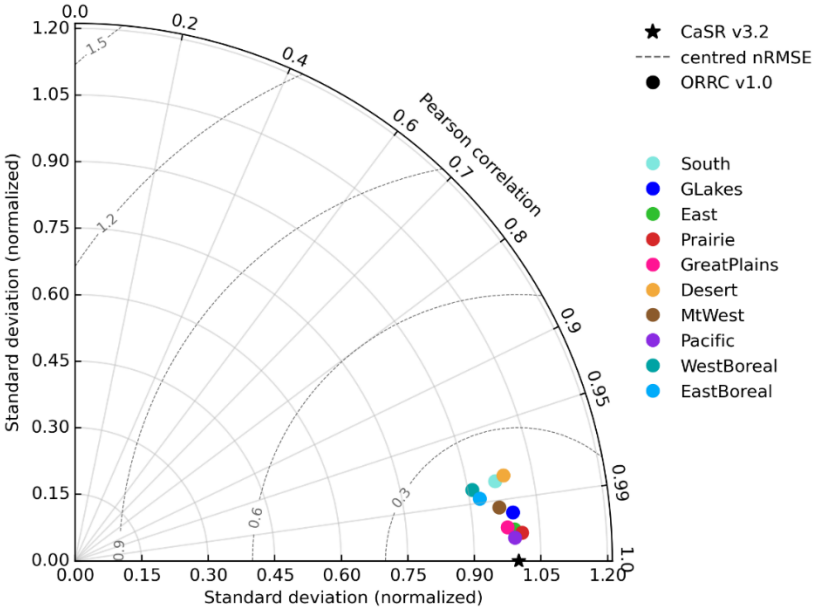


Figure 4. Taylor diagram comparing ORRC v1.0 and CaSR v3.2 monthly mean precipitation over 2020-2024 for the selected Bukovsky regions. For each region, the statistics are computed from the time series of spatially averaged monthly precipitation. Radial distance indicates the normalized standard deviation, angle indicates the Pearson correlation coefficient, and dashed contours show the centred, normalized root-mean-square error (centred nRMSE). The star marks the CaSR reference and coloured circles denote the individual Bukovsky regions.

Spatial differences are evaluated seasonally at the grid-cell scale from monthly means (Figure 5). The root-mean-square error (RMSE) relative to CaSR is computed from the monthly differences within each season. It is normalized by the standard deviation of the CaSR monthly anomalies (i.e., departures from the monthly climatology) to obtain the normalized RMSE (nRMSE). This normalization places the differences between the datasets in the context of CaSR's interannual variability, allowing comparisons across regions and seasons with different levels of variability. Differences are relatively limited over much of the continental interior but become larger and more spatially heterogeneous toward northern Canada, the Arctic, and parts of the southern domain. The contrast is most pronounced in in the summer, when elevated nRMSE becomes more widespread, including over eastern Canada. The magnitude of nRMSE is also influenced by the local variability of the CaSR reference field. Grid cells where the CaSR standard deviation is below the 5th percentile are therefore hatched. In these regions, even modest absolute differences can produce large normalized errors, and the highest nRMSE values should consequently be interpreted with caution. Corresponding spatial maps for the remaining variables and for ORRC-a v1.0 are available in the ORRC GitHub repository.

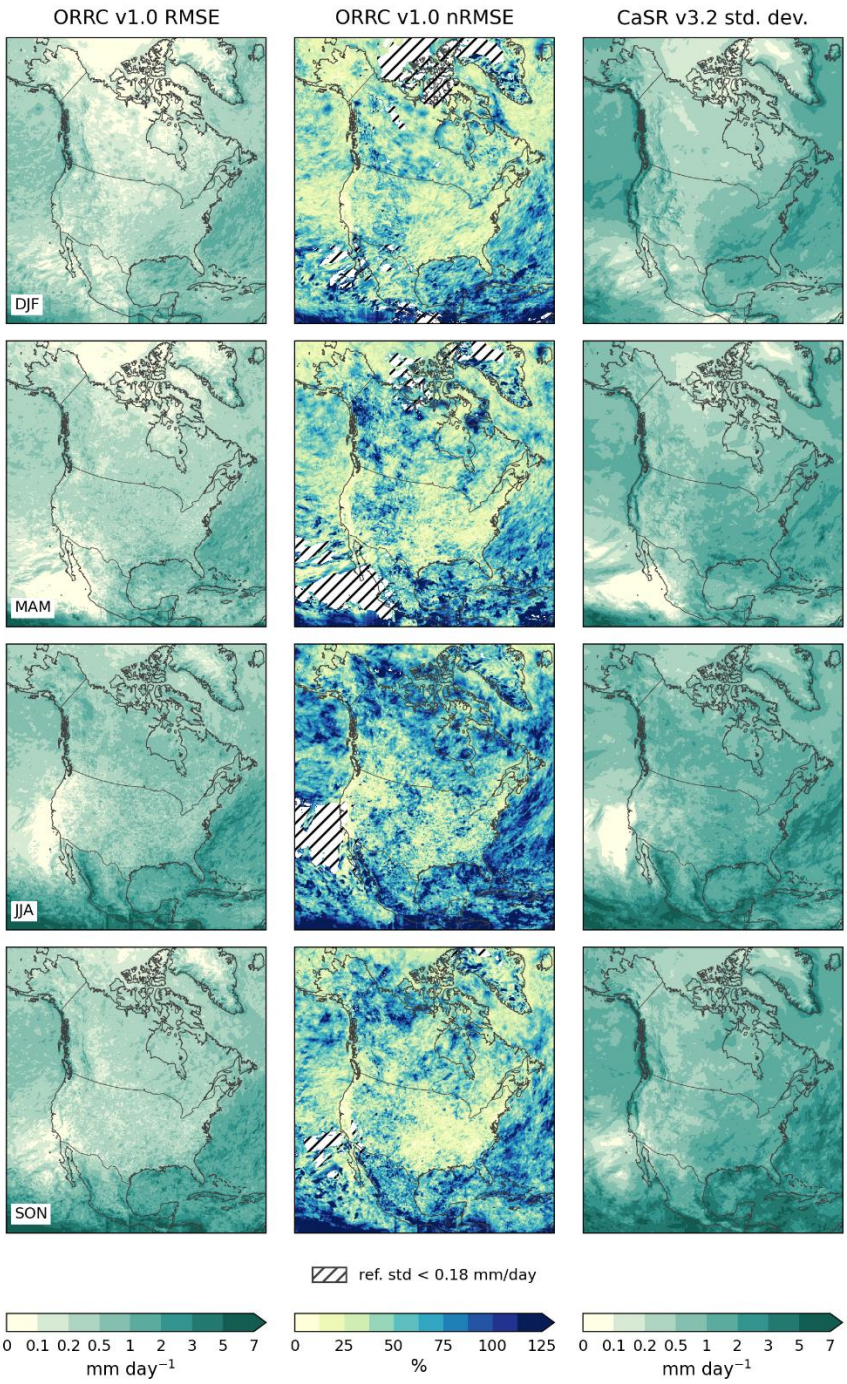


Figure 5. Seasonal agreement between ORRC v1.0 and CaSR v3.2 for precipitation, based on monthly means over 2020-2024. Rows show winter (DJF), spring (MAM), summer (JJA), and autumn (SON). The left column shows the root-mean-square error (RMSE; mm day^{-1}) of ORRC relative to CaSR. The right column shows the standard deviation of CaSR monthly anomalies (interannual variability; mm day^{-1}). The centre column shows the normalized RMSE (nRMSE; %), the RMSE divided by the standard deviation; values near or above 100% indicate that differences between the products are comparable to or larger than CaSR interannual variability. Hatching marks cells where the CaSR standard deviation is below 0.18 mm day^{-1} which corresponds to the 5th percentile of non-zero values pooled across seasons and grid cells. In these cells the small denominator can inflate the nRMSE.

Seasonal mean bias maps are used to examine systematic differences and assess the effect of the bias adjustment. Because ORRC-a is adjusted over the full 2020-2024 period, its agreement with CaSR over this period is an in-sample result and is expected to be close by construction. To assess the independence between the bias adjustment and evaluation periods, a cross-validated version of ORRC-a is produced, in which each year of 2020-2024 is held out in turn. For each held-out year, the bias adjustment is trained over the remaining four years (e.g., 2020 is excluded and the adjustment is trained over 2021-2024) and then applied to the held-out year. The five independently adjusted years are concatenated to form a continuous 2020-2024 dataset, which is evaluated against CaSR in the same way as ORRC and ORRC-a. This dataset is used only for this assessment; the released ORRC-a dataset is adjusted over the full 2020-2024 period in order to maximize the data available for training.

Figure 6 shows the seasonal mean precipitation bias relative to CaSR v3.2 over the domain for ORRC, ORRC-a adjusted over the full 2020-2024 period and ORRC-a v1.0 produced with cross validation. The bias adjustment substantially reduces the magnitude and spatial extent of precipitation biases in all seasons, and the cross validated results demonstrate the generalizability of the ORRC-a extension, lending support to its reliability beyond the end of the CaSR period in 2024. Additional maps showing the effect of the bias adjustment for 1.5 metre mean, minimum and maximum temperature are available in the ORRC GitHub repository.

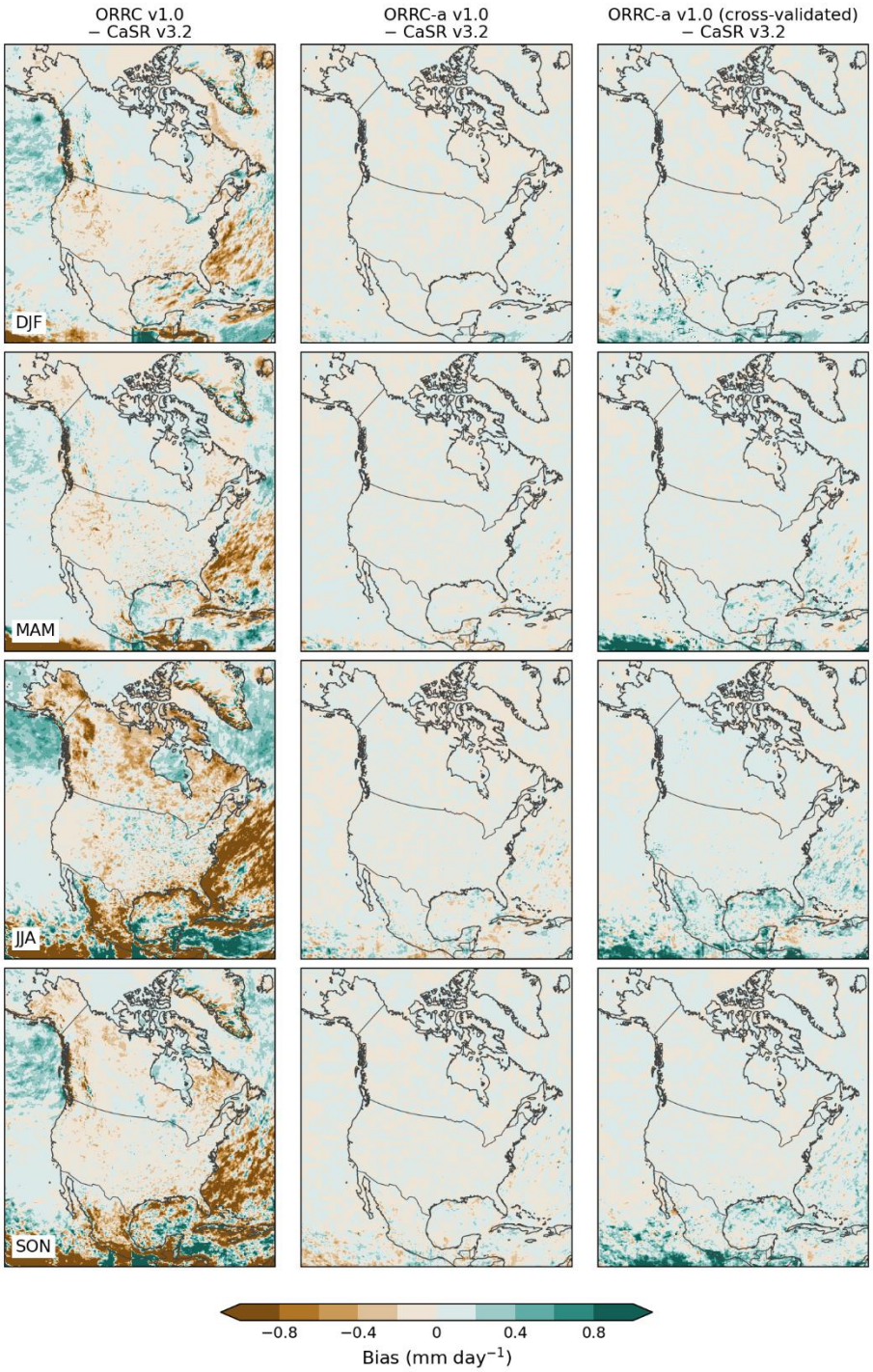


Figure 6. Seasonal precipitation bias (mm day^{-1}) relative to CaSR v3.2 over 2020-2024, computed as the difference between the seasonal mean of each product and that of CaSR v3.2. Columns show ORRC v1.0 (left), ORRC-a v1.0 adjusted over the full 2020-2024 period (centre), and ORRC-a v1.0 produced with cross validation (right), in which each year is adjusted using a bias correction trained on the other four years. Rows show winter (DJF), spring (MAM), summer (JJA), and autumn (SON).

References

Bourgault, P., Huard, D., Smith, T. J., Logan, T., Aoun, A., Lavoie, J., Dupuis, É., Rondeau-Genesse, G., Alegre, R., Barnes, C., Laperrière, A. B., Biner, S., Caron, D., Ehbrecht, C., Fyke, J., Keel, T., Labonté, M.-P., Lierhammer, L., Low, J.-F., Quinn, J., Roy, P., Squire, D., Stephens, A., Tanguy, M., and Whelan, C. (2023). xclim: xarray-based climate data analytics, *Journal of Open Source Software, 8*(85), 5415. https://doi.org/10.21105/joss.05415

Bourgault, P., Dupuis, É., Smith, T. J., Aoun, A., Beaupré Laperrière, A., Braun, M., Gammon, S., Huard, D., Lamarche, A., Lavoie, J., Rondeau-Genesse, G., & Roy, P. (2026). *xsdba: Statistical downscaling and bias adjustment library* [Computer software]. Zenodo. https://doi.org/10.5281/zenodo.18613881

Bukovsky, M. S. (2011). *Masks for the Bukovsky regionalization of North America*. Regional Integrated Sciences Collective, Institute for Mathematics Applied to Geosciences, National Center for Atmospheric Research. http://www.narccap.ucar.edu/contrib/bukovsky/

Dimetrijevic, M. (2023). *dimmilena/rdpa_as_rdrs* [Python]. https://github.com/dimmilena/rdpa_as_rdrs (Original work published 2022)

Environment and Climate Change Canada. (2019, July). *Changes to the Regional Deterministic Prediction System (RDPS) from version 6.0.0 to version 7.0.0* (RDPS Version 7.0.0 Technical Note). Canadian Meteorological Centre. https://collaboration.cmc.ec.gc.ca/cmc/cmoi/product_guide/docs/tech_notes/technote_rdps-700_20190703_e.pdf

Environment and Climate Change Canada. (2024a). *Version 6.0.0 of the Meteorological Service of Canada (MSC) Regional Deterministic Precipitation Analysis (RDPA) system*. https://collaboration.cmc.ec.gc.ca/cmc/cmoi/product_guide/docs/tech_specifications/tech_specifications_RDPA_e.pdf

Environment and Climate Change Canada. (2024b). *The Regional Deterministic Prediction System (RDPS) version 9.0.0 of the Meteorological Service of Canada (MSC)*. Meteorological Service of Canada, Canadian Meteorological Centre. https://collaboration.cmc.ec.gc.ca/cmc/cmoi/product_guide/docs/tech_specifications/tech_specifications_RDPS_e.pdf

Fortin, V., Roy, G., Stadnyk, T., Koenig, K., Gasset, N., & Mahidjiba, A. (2018). Ten Years of Science Based on the Canadian Precipitation Analysis: A CaPA System Overview

and Literature Review†. *Atmosphere-Ocean*, *56*(3), 178–196. https://doi.org/10.1080/07055900.2018.1474728

Gasset, N., Fortin, V., Dimitrijevic, M., Carrera, M., Bilodeau, B., Muncaster, R., Gaborit, É., Roy, G., Pentcheva, N., Bulat, M., Wang, X., Pavlovic, R., Lespinas, F., Khedhaouiria, D., & Mai, J. (2021). A 10 km North American precipitation and land-surface reanalysis based on the GEM atmospheric model. *Hydrology and Earth System Sciences*, *25*(9), 4917–4945. https://doi.org/10.5194/hess-25-4917-2021

Khedhaouiria, D., Gasset, N., Fortin, V., Dimitrijevic, M., Bulat, M., & Wang, X. (2026). The Canadian Surface Reanalysis (CaSR) v3.2 precipitation dataset: A 45-year high-resolution analysis for North America (1980–2024). *Hydrology and Earth System Sciences, 30*, 5971–5998. https://doi.org/10.5194/egusphere-2026-620

Mai, J., Kornelsen, K. C., Tolson, B. A., Fortin, V., Gasset, N., Bouhemhem, D., Schäfer, D., Leahy, M., Anctil, F., & Coulibaly, P. (2020). The Canadian Surface Prediction Archive (CaSPAr): A Platform to Enhance Environmental Modeling in Canada and Globally. *Bulletin of the American Meteorological Society*, *101*(3), E341–E356. https://doi.org/10.1175/BAMS-D-19-0143.1

Taylor, K. E. (2001), Summarizing multiple aspects of model performance in a single diagram. *Journal of Geophysical Research: Atmospheres, 106*(D7), 7183–7192. https://doi.org/10.1029/2000JD900719